

Lab 2 Requirements

1. Create a new database in MySQL that consists of the ten Euphoric State relations. Import the tables directly from the source data tables.
2. **Edit the design of your tables:**
 - Check that each field has an appropriate data type and size. Unsigned numeric data types should be used for numeric fields that cannot have negative values. The fields year, rank and letter should have an enum datatype.
 - Create a *primary key* for each table
 - Decide the fields where values should be mandatory (not null).
 - Create the relationships between your tables with *foreign keys*
3. **Create the following queries in SQL:**
 1. How many courses does each department gives? Include departments (if any) that does not give any courses
 2. List the teachers that does not teach this semester
 3. List the members in the committee "Course Planning" (name, department)
 4. List the teachers that are responsible for more than two courses
 5. List the teachers that **can** teach the course "Linear Algebra"
 6. List, for each student, the number of course the student takes.
 7. List for each course in Computer Science that has more than four students: the name of the course, the highest grade, the lowest grade, and the average grade on the course.
 8. List name and (numeric) grade for the students on the course "CS Introduction" that have received the grade 'C'.
 9. List for each course in Mathematics: the name of the course, the number of students and the teacher that does teach the course
 10. List for each student that has failed a course: the student's name, the name of the course and the grade

4. Create the following triggers in SQL:

- A trigger for UPDATE of the field grade: if the new value > 100 an error message should be printed
- A trigger for INSERT of a new course: if does_teach and responsible are not employed at the same department an error message should be printed
- A trigger for DELETE of a student: the deleted student should be added to the table Alumni_students (which you create in advance)

To submit

See the documentation template

Requirements for approval

All SQL queries and triggers should be tested by a supervisor before submission.